

EDA

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Clean the Data

```
library(dplyr)
library(tidyverse)
library(ggplot2)
library(tsibble)
library(feasts)
library(fable)
```

```
singapore <- read.csv('BirthsAndFertilityRatesAnnual.csv')
```

```
clean_1 <- singapore |>
  filter(DataSeries %in% c("Total Fertility Rate (TFR)", "Total Live-Births"))
```

```
clean_2 <- pivot_longer(clean_1,
  cols = starts_with("X"),
  names_to = "Year",
  values_to = "Value",
  values_transform = list(Value = as.character))
```

```
clean_3 <- mutate(clean_2, Year = str_remove(Year, "X"))
```

```
clean_4 <- clean_3 |> mutate(Year = as.numeric(Year), Value = as.numeric(Value))
```

```
clean_data <- clean_4 |>
  pivot_wider(names_from = DataSeries, values_from = Value)
singapore <- clean_data |>
  as_tsibble(index = Year)
singapore
```

```
## # A tsibble: 66 x 3 [1Y]
##   Year 'Total Fertility Rate (TFR)' 'Total Live-Births'
##   <dbl> <dbl> <dbl>
## 1 1960 5.76 61775
## 2 1961 5.41 59930
## 3 1962 5.21 58977
## 4 1963 5.16 59530
## 5 1964 4.97 58217
```

```
## 6 1965          4.66          55725
## 7 1966          4.46          54680
## 8 1967          3.91          50560
## 9 1968          3.53          47241
## 10 1969         3.22          44562
## # i 56 more rows
```

```
singapore |> autoplot(`Total Fertility Rate (TFR)` ) +
  labs(title = "Total Fertility Rate (1960-2024)",
        caption = "Fig. 1 Plot of the Total Fertility Rate in Singapore from 1960 to 2024.",
        y = "Total Fertility Rate",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Warning: 'autoplot.tbl_ts()' was deprecated in fabletools 0.6.0.
## i Please use 'ggtime::autoplot.tbl_ts()' instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
##   'library(ggtime)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

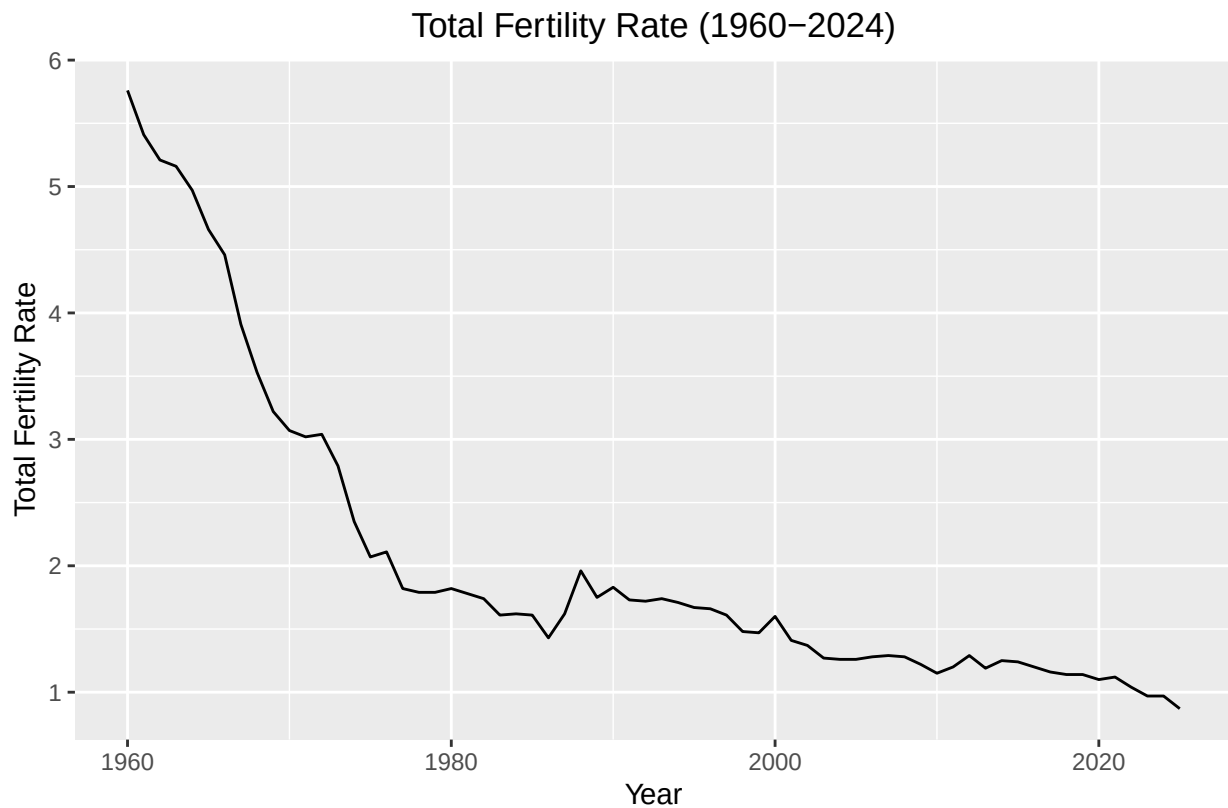


Fig. 1 Plot of the Total Fertility Rate in Singapore from 1960 to 2024.

```
singapore |> autoplot(`Total Live-Births`) +
  labs(title = "Total Live-Births (1960-2024)",
        caption = stringr::str_wrap("Fig. 2 Plot of the Total Live-Births in Singapore from 1960 to 2024"),
        y = "Total Live-Births",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

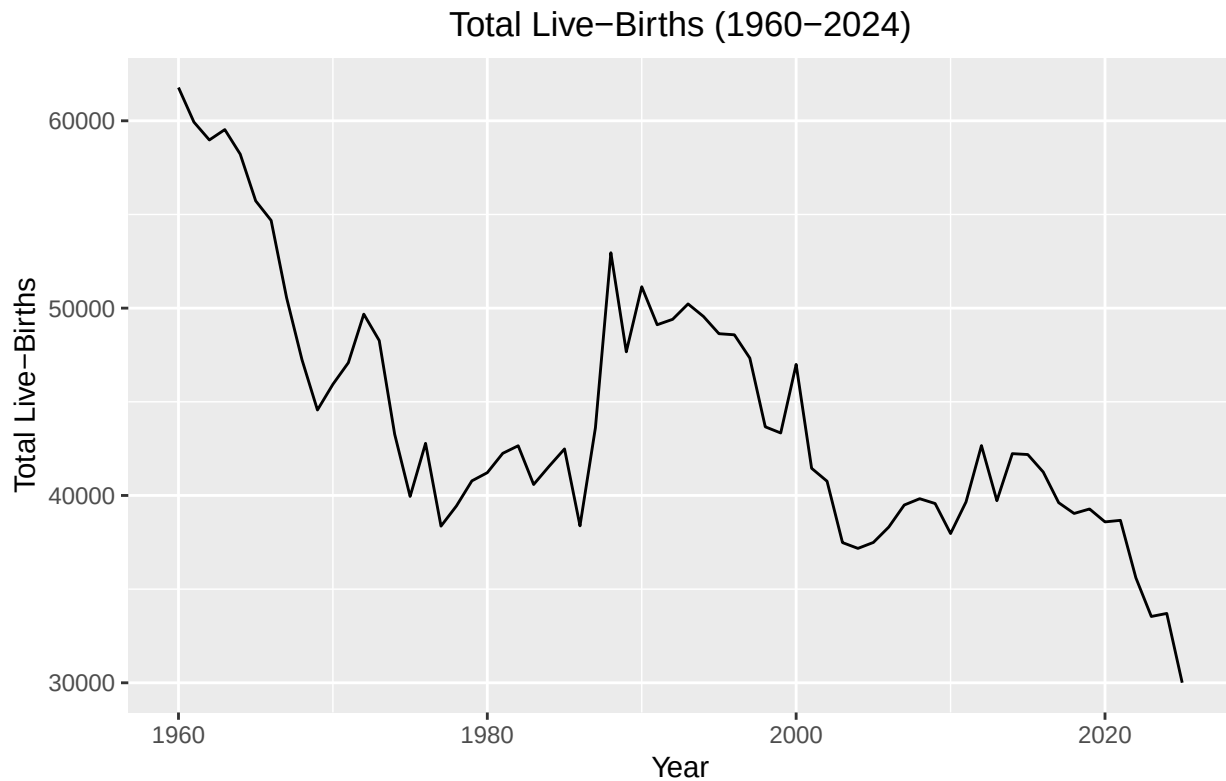


Fig. 2 Plot of the Total Live-Births in Singapore from 1960 to 2024 shows that it follows a roughly similar trend as the plot of the Total Fertility Rate.

While figure 1 and figure 2 look different we can see that they show spikes and drops at similar points. There is a spike around 1988 in the plot of Total Fertility Rate and a large spike around the same year in the plot of the Total Live-Births. This can also be seen with a spike around 1972, as well as drops around 1969, 1987, 2003, and 2024.

```
singapore |> autoplot(log(`Total Fertility Rate (TFR)`)) +
  labs(title = "Log-Transformed Total Fertility Rate (1960-2024)",
        caption = stringr::str_wrap("Fig. 3 Plot of the Total Fertility Rate with a log transformation in"),
        y = "log(Total Fertility Rate)",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

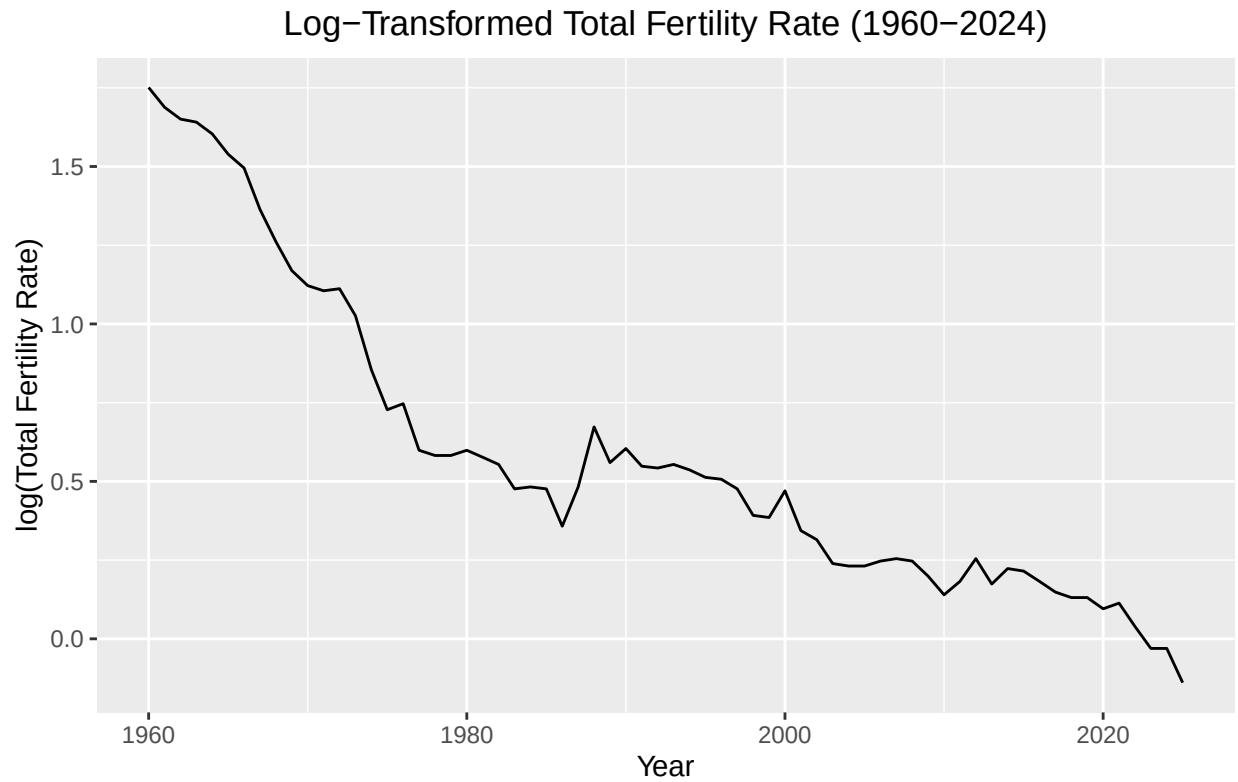


Fig. 3 Plot of the Total Fertility Rate with a log transformation in Singapore from 1960 to 2024.

```
singapore |> autoplot(log(`Total Live-Births`)) +
  labs(title = "Log-Transformed Total Live-Births (1960-2024)",
        caption = stringr::str_wrap("Fig. 4 Plot of the Total Live-Births with a log transformation in S",
        y = "log(Total Live-Births)",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

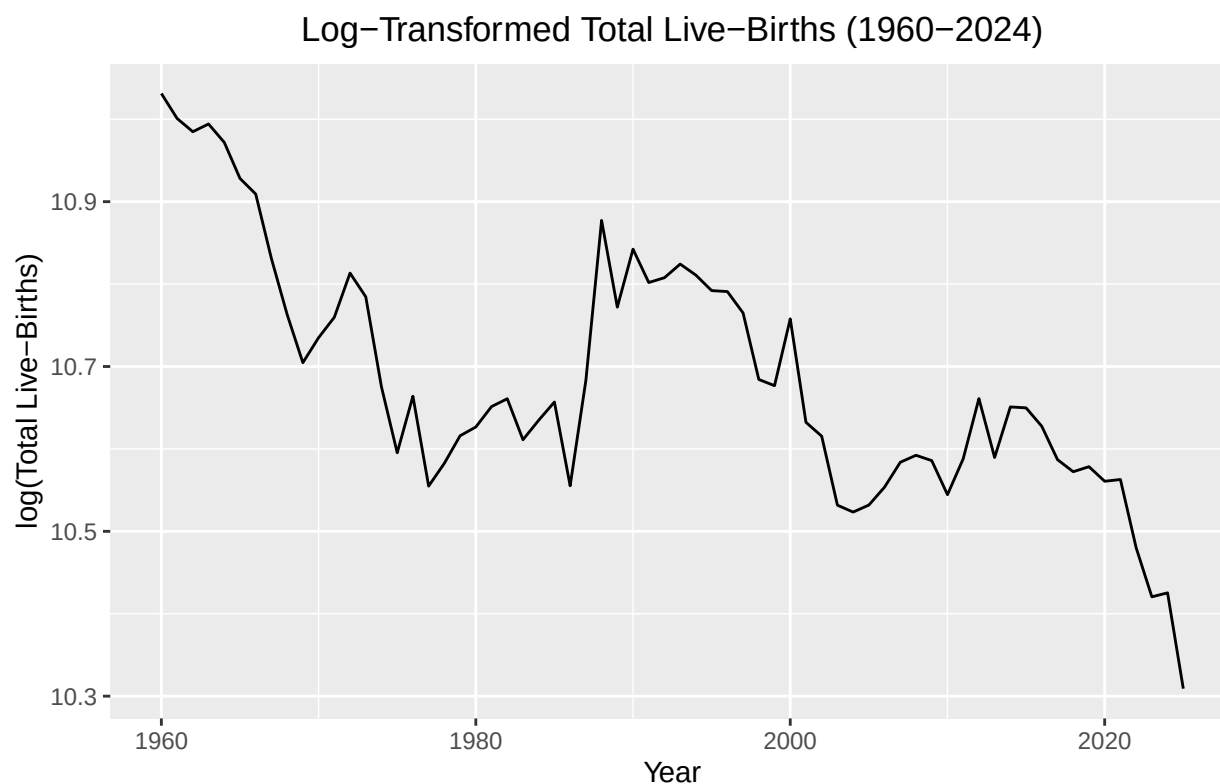


Fig. 4 Plot of the Total Live-Births with a log transformation in Singapore from 1960 to 2024.

```
singapore |> features(log(`Total Fertility Rate (TFR)`), unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>    <dbl>
## 1     1.49      0.01
```

Table 1. KPSS test of the log transformed Total Fertility rate in Singapore from 1960-1924. The p-value is 0.01, which is significant at the $\alpha = 0.05$ significance level.

```
singapore |> features(log(`Total Live-Births`), unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>    <dbl>
## 1     0.996      0.01
```

Table 2. KPSS test of the log transformed Total Live-Birth rate in Singapore from 1960-1924. The p-value is 0.01, which is significant at the $\alpha = 0.05$ significance level.

Figures 3 and 4, along with the p-values from tables 1 and 2 both being significant at the 0.05 significance level, show that a log transformation seems to help a bit but they both still need to be stabilized more. To do this, they will be differenced.

```
singapore |> autoplot(difference(log(`Total Fertility Rate (TFR)`), 1)) +
  labs(title = "First-Order difference of Log TFR (1960-2024)",
        caption = stringr::str_wrap("Fig. 5 Plot of the first-order difference of the log transformed To",
        y = "Change in log(TFR)",
        x = "Year")) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').
```

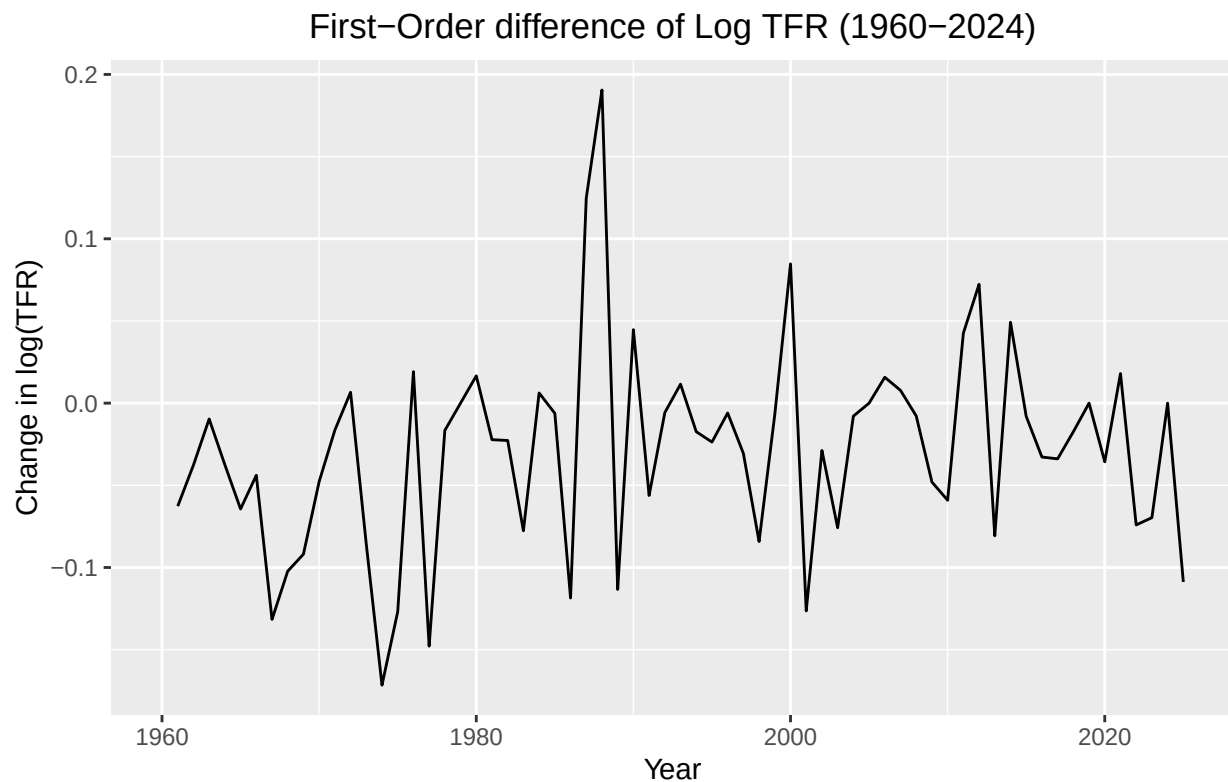


Fig. 5 Plot of the first-order difference of the log transformed Total Fertility Rate in Singapore from 1960 to 2024.

```
singapore |> autoplot(difference(log(`Total Live-Births`), 1)) +
  labs(title = "First-Order difference of Log TLB (1960-2024)",
        caption = stringr::str_wrap("Fig. 6 Plot of the first-order difference of the log transformed To",
        y = "Change of log(TLB)",
        x = "Year")) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').
```

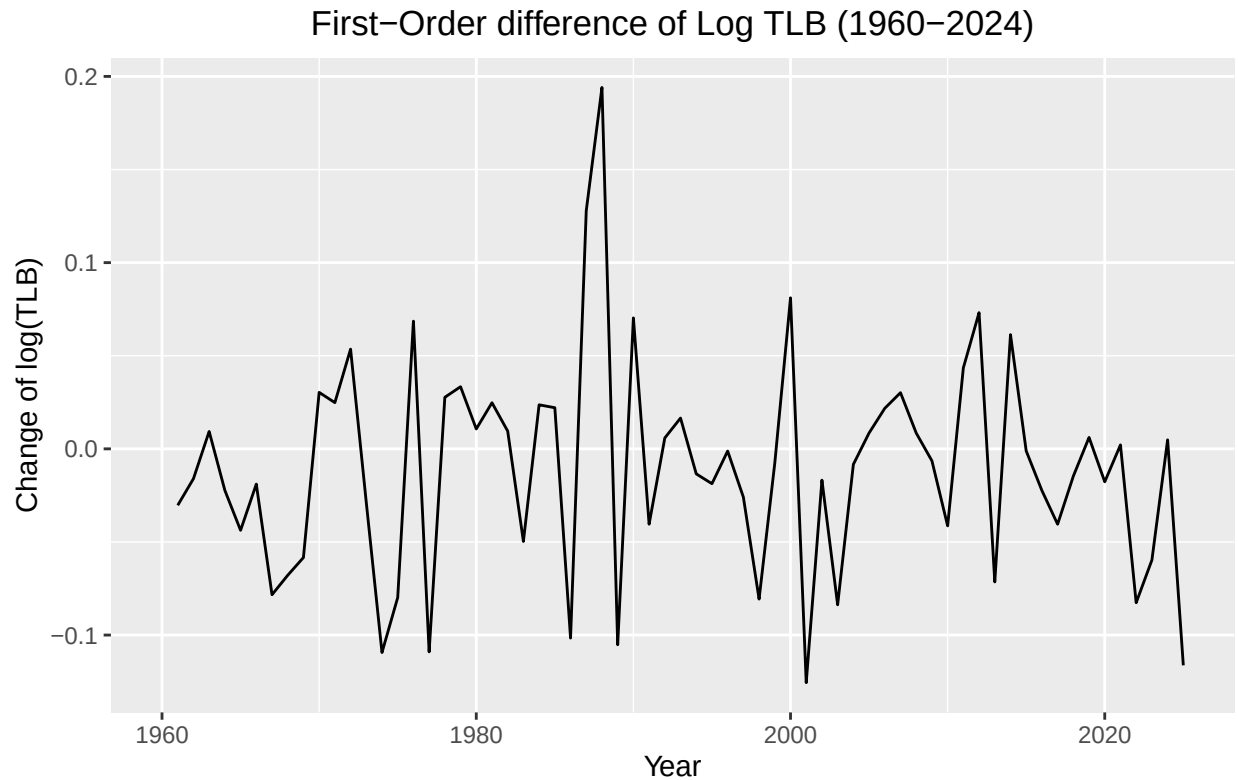


Fig. 6 Plot of the first-order difference of the log transformed Total Live-Births in Singapore from 1960 to 2024.

Plots 5 and 6 no longer show a downward trend with a first-order difference of the log transform of each variable.

```
TFR <- difference(log(singapore$`Total Fertility Rate (TFR)`), 1)
```

```
TLB <- difference(log(singapore$`Total Live-Births`), 1)
```

```
singapore |> features(TFR, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>      <dbl>
## 1     0.370      0.0902
```

Table 3. KPSS test of the first-order differenced log transformed Total Fertility rate in Singapore from 1960-1924. The p-value is 0.09, which is not significant at the $\alpha = 0.05$ significance level.

```
singapore |> features(TLB, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>      <dbl>
## 1     0.114      0.1
```

Table 4. KPSS test of the first-order differenced log transformed Total Live-Birth rate in Singapore from 1960-1924. The p-value is 0.1, which is not significant at the $\alpha = 0.05$ significance level.

Tables 3 and 4 confirm what figures 5 and 6 showed, when both Total Live-Births and Total Fertility rate are log transformed and then differenced, they are stationary. In a KPSS test, if the p-value is less than 0.05, the null-hypothesis that the series is stationary, is rejected. However, if the p-value is greater than 0.05, the null-hypothesis is accepted and the series is found to be stationary.